Variance explained by PCA

Frozen 1B LSTMs · means across 8 seeds · varying movement and goals

Measure	Grid	PGC + grid
Variance captured by first 3 PCs	23.8%	29.4%
Variance captured by first 10 PCs	46.4%	52.3%
PCs needed for 50% variance	12.4	9.5
PCs needed for 60% variance	20.0	15.3
PCs needed for 70% variance	31.1	25.4
PCs needed for 80% variance	49.4	43.3
PCs needed for 90% variance	82.0	77.5
PCs needed for 95% variance	112.9	109.6
PCs needed for 99% variance	164.9	165.5
Full PCA basis (100% variance)	256.0	256.0

PC = principal component. Percentages show variance captured; other values are component counts.

Counts are averaged across seeds, so they can be fractional. 100% uses the full 256-PC basis.

Explained variance measures variation in hidden activity, not task accuracy.